1. What is website?

A website is a collection of information and related web pages connected through links. It is accessible through the internet using web browsers generally every website have a unique domain name and is used to share information, communicate, purpose products, watch videos and provide online services is developed using HTML, CSS, and JS.

2. What is Web page?

A Webpage is a simple single document or individual page on the internet. It is a part of a website access through the web browsers. a web page is created to display specific information to users It can contain text, images, videos, audios, links buttons and forms multiple related web pages are connected through hyperlinks and navigation menus to form a complete website.

3. What is an attribute in HTML? Explain its type with example.

Attribute is a special property used to provide additional information about an HTML element. Attributes define the behaviour, appearance, identity and functionality of HTML elements. it contains two parts name and value attribute = "value" attributes helps developers control how HTML element works and display content.

1. Global Attribute - <h1 id="" class="">
2. Specific Attribute - <img src="", alt="" width="">
3. Boolean Attribute - <input type="" required>
4. Event Attribute - <button onclick="">

4. What is CSS layout? Explain different CSS layout methods and their uses.

CSS is a technique used to arrange and organize HTML element on a web page. It controls how elements are positioned, aligned, size and displayed inside a webpage. CSS layout helps developers create structured, attractive and user friendly website designs. It helps to develop responsive websites that adjust to different screen sizes, such as mobile phones, tablets and desktop.

1. Normal flow layout - Block elements and Inline elements.
2. Flexbox layout - It allows a row and column.
3. Grid layout - CSS is control both horizontal and vertical
4. Position layout - static, relative, absolute, fixed, sticky
5. Float layout - position the element to left or right of the container.
6. Multi column layout - column-count, column-gap, column-rule.

5. What is Programming language?

A. A Programming language is a formal language used to communicate instructions to a computer. It allows developers to create programs, applications, websites, and software systems. It is used to provide instructions to a computer to perform specific tasks. is help automate manual tasks and solve real-world problems. used in mobile application, artificial intelligence, machine learning, and software development - programing language have its own rules and structure. known as syntax.

6 What is JS and it features?

JavaScript is a high-level programming language used to make websites interactive and dynamic. It helps developers add functionality to websites instead of displaying only static content. It responds the user action such as clicking buttons, entering texts, submitting forms, and selecting options. It also communicate with server and database through APIs. It works along with the HTML and CSS to create interactive websites.

1. Light weight - ~~one~~
2. Interpreted language
3. Dynamic Typing
4. Object-based language
5. Event Handling.
6. Platform Independent
7. Asynchronous Programing
8. DOM Manipulation.

7,8. What is function in JavaScript? Explain its types.

A function in JavaScript is a block of reusable code designed to perform a specific task. function helps to us organize code into smaller and manageable parts. ~~because~~ a function executed when it is called or invoked. it accepts input values called parameter. process the input and produce an output using the return statement. It makes the program ~~so~~ easier to understand, maintain and debug.

1. Named function - Is defined with a specific name.
2. Anonymous function - A function without a name is called anonymous fun
3. Arrow function - shorter syntax for writing functions.
4. Funtion with parameter - Parameters are values passed by function.

5. Funtion with return value - return statement sends a result back.
6. Immediately Invoked function expression -
7. Callback function - It pass argument to other function.

8. What is fetch API in JS. Explain how its data using fetch.

The fetch API is built in JavaScript ~~function~~ feature used to request data from a server. It allow JS to communicate with external API and web servers and display it on a web page. It return Promises which helps handle asynchrous operation. handle response .then() and catch()
It is works with JSON data, commonly in backend.
Using Fetch API:
send a request using fetch()
converts the response JSON using .json().
Display and use the recive data.
Asyn and Await proces. .then() and .catch()

9.

9. What is React js? Explain its features.

A: React js is a JavaScript library used to build user interfaces (UI) for websites and web applications. It was developed by facebook now (meta). It helps developers to create interactive and dynamic webpage. to divide a website into small, reusable parts and components. React use js and jsx to create a interface. It is comonly used to build single-application.
features:-

1. Component-based Architecture
2. Virtual Dom
3. JSX (JavaScript)
4. Reusable Components.
5. One-Way data Binding
6. Props
7. State Managment
8. Hooks.

10. What is ES6 and ES6+ in JavaScript?

A. ES6 stands for ECMA Script 2015. It is a major version of ECMA ~~Javascript~~ standard. It defines how js should work. ES6 was introduce in 2015 how javascript works. introduce new features that made JavaScript code shorter, clener, and easier to maintain. It introduce arrow funtion, template literals, destructing and many other funtion.

ES6+

It refer to ES6 and new feature introduced in JavaScript after ES6. It includes improvements introduced in later ECMA script version. These features make JS more powerfull and easier to use. write clean and reusable code. It includes. features. optional chaining, nullish collescing, async/await and more. developers normally used ES6+ features.

11. What is differnce between NPM and NPX.

NPM

NPM stands for Node Package Manager. It is install, manage, update ~~Manager~~. and remove js package. it is a reusable peace of code developed by other developers. npm install libraries. tool, React, Express, Bootstrap and Axios. It manage project using package.json file. . npm install package into node_modules installing locally.

NPX?

NPX stands for Node Package Execute It is a command line tool that comes with modern npm version. It is execute package globally. download the execute package temporarily when needed. It is commonly used to create React and other JS projects

12. What is Babel, esbuild, and Webpack? Explain bundlers.

~~What is a Bundler~~

A bundler is a development tool that combines a multiple files using in a project into one or more optimized output files. A bundler collects the required files and organizes them for the browsers. It analyse imports and export.

Babel is js compiler (transpiler). some older browser may not support all modern Javascript features. Babel transform support to modern syntax into compatiable Javascript syntax.

Esbuild:-

Esbuild is a fast Javascript and CSS bundling tool. It is written in go program language. It is a modern frontend tool. including uite, It supports multiple files to optimized output files.

Web-pack:-

Is a model bundler. It collect the different files from the project and combines their dependance into output bundles. It uses plugins to perform additional tasks, It support code splitting.

13. What is Vite? Explain its uses and features.

Vite is modern frontend build tool used to develop and build web applications. It is created by Evan You, who also created the.js. It provides a fast development server for frontend projects. It helps develop, start their project quickly. vite use Native ES modules during development. It use Rollup for production bundling. It used in modern React and frontend development.

Features:

Fast development server
Hot Module Replacement
Easy Project setup
Support React
Fast Build process
CSS supports.
Type-script support
Asset handling
Environment variables.

14. What is Oxc, Rollup and Roll down?

Oxc stands for Oxidation Compiler. It is a collection of high-performance JavaScript and Typescript tools. It is written primarily in the Rust Programming language. It develops the identify code errors and maintain code quality.

Rollup is a Module bundler. it collects multiple JavaScript files and their dependencies into one or more output files. It supports Tree Shaking, which removes unused code from the bundle of final. it makes flexible and optimized production builds.

Roll down is a JavaScript and Typescript bundler written in Rust fast. High performance bundling. It is now the bundler used by Vite & inspired by the Rollup and esbuild.

15. NPM Run dev, npm start, and npm run build? Explain their purpose:

NPM Run dev

npm run dev is used to start the development server while writing and testing code. It runs your project on a local development server. It has a HMR.

NPM start :-

npm start executes the start script defined in package.json to run an application. It is used in often Node.js backend applications.
It execute the cmd under the start script.

NPM run build :-

npm run build generates optimized production files that are ready for deployment. It convert process to source files. It bundle the required files. It generates production-ready output files.

16. What is JSX ? What is the difference between JSX and JavaScript ?

JSX is stands for JavaScript XML

JSX is syntax extension used in React that allows developers to write HTML-like code inside Javascript. It helps to create the user interface of React application in HTML like syntax. JSX transform into JavaScript before execution.

JSX	JavaScript
It is a JavaScript XML.	It is a Programming Language.
Uses HTML-like syntax.	Uses JavaScript syntax
Need to transformation before browser can execute it.	Browser can execute JavaScript directly.
Allows HTML-like elements inside JavaScript.	Does not directly support HTML-like syntax.
React describe the UI	It logical and functionality.